

Finite Tetrahedral Complexes, Skew Support Flow and Contact Readiness

Paper I - Consolidated integration draft 1.1

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3 October 2026

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Edition status. Integration draft for independent review. Source baseline: portal version 161, commit 62a7c594e0c7ab02f1c87702d0586a39352170d5. Mathematical assertions retain the premises of their declared carriers and maps. Source-audited calculations are distinguished from new derivations. Physical promotion: zero. Independent human review: pending.

Abstract

We consolidate finite tetrahedral chain algebra with ambient jump comparisons, signed seed readiness and supplied multi-copy contact dynamics. The unit-weight skew Hodge-Dirac baseline, matching-indexed binary carrier, unique-simplex ownership identities, exact A/B/C spectra and receipt-conditioned affine maps are retained. A common ambient comparison exhibits incompatible linear chain/isometry/boundary contracts. A fixed-coupling theorem distinguishes immediate, fourth-order and absent recipient population, while candidate face-readiness coefficients require prepared polarity. Reciprocal endpoint charts give conditional conservation and continuation on supplied contact graphs. Legal distinct-face contacts need not commute, and symmetry obstructs deterministic exclusive progress at a symmetric tie. These results separate carrier geometry, amplitude population, event receipts and admission. Historical constitutive, winding, clock, engineering and empirical updates are integrated with their source limits. No native geometry-creation law, physical coupling or empirical validation is inferred.

1. Scope and mathematical question

A finite complex, a linear evolution on its chain space, and a rule for changing the complex are different mathematical objects. This paper keeps them separate. It asks what follows exactly from a specified tetrahedral complex and a fixed skew generator, and what additional information is needed to define a discrete event map.

The starting point is conventional: oriented simplices form bases of real chain spaces; signed incidence defines the boundary; the boundary and its adjoint define a Hodge Laplacian. We use the skew combination of these maps to obtain an orthogonal flow. Norm preservation is a standard consequence of skew symmetry, not a new unification principle. The specific calculations concern a matching-indexed binary state, three finite glued complexes, and conditional maps determined by recorded finite data.

The notation is chosen to make each object recognizable before introducing any project terminology. A finite oriented tetrahedral simplicial complex is sometimes called a *packet complex* in the wider Unified Dynamics project. The Euclidean squared chain norm is denoted S_μ , or *support*. A *receipt* means only recorded finite data supplied to a stated map. None of these names changes the mathematical type of the object.

What this paper does not claim. The finite chain space is not identified with physical space, a physical field, matter or energy. The evolution parameter has no assigned physical unit. The binary coordinates are labels, not measured degrees of freedom. No microscopic source of the event receipts is derived. No physical coupling, action scale, experimental validation or universal recurrence is inferred from the calculations. A structural or numerical correspondence with another construction would not, by itself, identify the two objects.

The original result sequence is retained and extended. Section 3 defines the labeled state set and its symmetries. Section 4 fixes the chain convention and records standard operator identities. Section 5 establishes local norm balance and gluing accounting. Section 6 gives the exact A/B/C spectra, recurrence obstruction and ambient linear-jump incompatibilities. Section 7 is one conditional argument: given event receipts, the affine binary map is exact; the pre-event source of those receipts remains open. Section 8 now treats signed readiness, contact charts and admission. Appendix C retains the preservation distinctions. The claim table and executable checks provide a finite review surface.

Projective reconstruction is developed in the separate companion paper. Optical acquisition is a separate proposed protocol. Neither is required to prove a statement here.

2. Related work and the contribution boundary

Discrete exterior calculus, combinatorial Hodge theory, Dirac operators on complexes, lattice gauge theory and bistellar moves already supply much of the relevant mathematical language. They are the starting point, rather than discoveries of this paper.

Hirani's discrete exterior calculus [1] develops discrete forms and metric-dependent operators using primal and dual complexes. The present work uses a much narrower choice: an orthonormal simplex basis and ordinary transposes. It introduces neither a circumcentric Hodge star nor a convergence theorem for a continuum equation. The comparison is to the discrete incidence/adjoint construction; the thesis abstract and the author's overview [2, PDF pp. 1-3] identify this setting. We do not infer that the unit-weight matrices here approximate an externally specified metric.

Horak and Jost [3, arXiv version pp. 4-6, Definition 2.1] define the up, down and total combinatorial Laplacians using coboundaries and adjoints. Our block Laplacian is the unweighted, nonaugmented specialization of that familiar construction. Our calculation adds the explicitly listed spectra of the selected A/B/C complexes and their consequences for exact periods, not a new general Hodge theory.

Knill [4, pp. 1-2] describes $D = d + d^*$, its square, and the simplex-incidence graph. His Section 17, p. 18, also uses $d - d^*$ and its relation to the Laplacian. Accordingly, the skew operator and norm conservation below are standard setup. We apply them to a declared finite family and state exactly which inclusion is an isometry and which relabeling is a dynamical intertwiner.

Pachner's theorem [5] concerns equivalence of triangulations of piecewise-linear manifolds under elementary changes. Lickorish [6, p. 302, Definition 2.3; p. 315, Theorem 5.9] states the local bistellar replacement and the global equivalence theorem. We use the three-dimensional 2-to-3

replacement and verify its oriented boundary identity. We claim neither a new move nor the global equivalence theorem. Equality of the exterior boundary does not imply equality of the interior chain operator or its spectrum.

Wilson's lattice gauge theory [7, Section III, pp. 2448-2450] introduces gauge variables and gauge-invariant lattice constructions. A binary label permutation or a change of serialization forest is not thereby a lattice gauge field. There is no Wilson action or gauge-field dynamics in this paper. Kraus's operator formalism [8] is relevant only to the final distinction between an operator contraction and a norm-preserving map; no new dilation theorem is asserted. The cited article is the original operator-theoretic source, not a source for any physical interpretation of our chain coefficients.

The reviewable differences from these literatures are therefore restricted to the following constructions and calculations:

1. the explicit opposite-edge indexing of $\Theta \cong \mathbb{F}_2^3$, distinguished from the matching stabilizer;
2. the selected unique-simplex A/B/C matrices, their exact spectra, and the irrational-frequency obstruction;
3. the receipt-conditioned affine Θ -map and its exact transition counts;
4. a typed comparison of Euclidean norm, a second quadratic form, sector mixing and contraction effects.

The companion paper treats the reconstruction statistic, its parity-degree ambiguity and the frozen-tree covariance map. We do not assert a priority claim for general graph synchronization, linear covariance propagation, or sufficient data for sign reconstruction.

3. Labeled tetrahedral states

3.1 Vertices, edges and perfect matchings

Let $V = \{0, 1, 2, 3\}$ and let K_4 be the complete graph on this vertex set. Its six edges are the two-element subsets of V . Every permutation of V preserves adjacency, so $\text{Aut}(K_4) = S_4$.

The three perfect matchings are

$$M_1 = \{01, 23\}, \quad M_2 = \{02, 13\}, \quad M_3 = \{03, 12\}.$$

Each partitions the four vertices into two disjoint edges. A vertex permutation induces a permutation of this three-element set. Write

$$\pi : S_4 \longrightarrow S_3$$

for that action.

Proposition P3.1 (matching action). The map π is onto and has kernel

$$V_4 = \{1, (01)(23), (02)(13), (03)(12)\}.$$

Proof. Each listed double transposition preserves all three matchings. A transposition of vertices fixes one matching and exchanges the other two, and suitable vertex permutations therefore generate all permutations of the three matchings. The kernel of a surjection from a group of order 24 onto one of order 6 has order 4. The four listed elements exhaust it. $\square$

Assign one bit to each matching:

$$\Theta = \mathbb{F}_2^{\{M_1, M_2, M_3\}} \cong \mathbb{F}_2^3.$$

For $p \in S_3$, use the coordinate action $(P_p \theta)_{p(j)} = \theta_j$. This convention fixes the direction of later composition formulas.

Proposition P3.2 (equal order is not group identity). The stabilizer of one matching in S_4 has order 8 and is nonabelian. It is not the additive group of Θ .

Proof. Orbit-stabilizer gives order $24/3 = 8$. In the stabilizer of M_1 , independently swap the two vertices of each edge, and also exchange the edges. Thus the group is $(C_2 \times C_2) \rtimes C_2$, with the last factor exchanging the first two. For example, the within-edge swap (01) and edge exchange (02)(13) do not commute. The additive group $\mathbb{F}_2^3$ is abelian. $\square$

The distinction is structural, not merely terminological. A matching indexes a coordinate; its stabilizer is a group of vertex permutations. The coordinate space is not obtained by identifying those two objects.

3.2 The edge projection and its fibers

Define the labeled state set

$$\mathcal{X} = S_4 \times \Theta.$$

This is a product of sets. It has 192 elements. To make a fiber count meaningful, a projection must be declared. Fix the reference edge $e_0 = \{0, 1\}$ and define

$$p_E(g, \theta) = g(e_0).$$

The second coordinate does not enter this projection.

Proposition P3.3 (uniform edge fibers). Every edge has 32 inverse images under p_E . Consequently

$$|\mathcal{X}| = 192 = 6 \times 32.$$

Proof. A permutation sending e_0 to a specified edge can assign its two vertices in two orders and assign the two complementary vertices in two orders. There are four such permutations. Each can be paired with any of the eight binary vectors, giving $4 \times 8 = 32$ states. The six fibers partition $\mathcal{X}$. $\square$

The integer factors are $192 = 2^6 \cdot 3$, $6 = 2 \cdot 3$, and $32 = 2^5$. They record the specified counting construction, not a dynamical scale. In quotient/remainder notation, $192 = 6(32) + 0$ and $192 \equiv 0 \pmod{32}$: six complete groups leave no remainder.

No-go NG3.4 (cardinality does not specify depth). The size 32 of a fiber does not imply 32 successive update stages, a 32-cycle, or a required duration.

Proof. On the same 32-element fiber one may define the identity map, a transposition, or a 32-cycle. Their update histories differ while the underlying cardinality is unchanged. No transition map is included in the fiber definition. Therefore no sequential depth follows from that definition alone. $\square$

3.3 Hamming sectors

On the eight-dimensional vector space with basis e_θ , define

$$W e_\theta = w_H(\theta) e_\theta, \quad w_H(\theta) = \theta_1 + \theta_2 + \theta_3.$$

These eigenspaces are called Hamming sectors; “branch” will not be used to imply a physical process.

Proposition P3.5 (sector dimensions). The four spectral projectors

$$P_w = \prod_{\substack{j=0 \\ j \neq w}}^3 \frac{W - jI}{w - j}, \quad w = 0, 1, 2, 3,$$

have ranks 1, 3, 3, 1. Coordinate permutations commute with W . Complementation $Qe_\theta = e_{\theta \oplus 111}$ satisfies $QWQ^{-1} = 3I - W$.

Proof. There are $\binom{3}{w}$ binary vectors of weight w . The displayed polynomials equal one at their designated eigenvalue and zero at the other three. Coordinate permutations preserve the number of ones; complementation replaces that number by three minus itself. $\square$

For any linear map U on this space, preservation of every Hamming sector is equivalent to $[U, W] = 0$. Indeed, the commutator vanishes exactly when all matrix blocks between different eigenvalues vanish. Equal dimensions of the weight-one and weight-two sectors do not choose a preferred sector.

4. Oriented chains and the standard skew generator

4.1 Chain convention

A simplicial complex in this paper contains each nonempty simplex once, including all faces of every maximal simplex. Increasing vertex order fixes a basis orientation. With real coefficients, let $C_k(K)$ be the chain space of k -simplices, and put

$$\mathcal{C}(K) = \bigoplus_{k=0}^3 C_k(K).$$

We use the nonaugmented complex: no empty-simplex coordinate is included. The boundary is

$$\partial_k[v_0 \dots v_k] = \sum_{j=0}^k (-1)^j [v_0 \dots \hat{v}_j \dots v_k].$$

Its matrix is B_k , with rows indexed by $(k-1)$ -simplices and columns by k -simplices. Inner products make these oriented basis vectors orthonormal. Adjoint therefore means transpose.

Lemma L4.1 (standard chain identity). For every such complex, $B_{k-1}B_k = 0$.

Proof. Every codimension-two face occurs twice in the iterated boundary. Removing vertices in the opposite order changes one deletion index by one, so the two signs are opposite. The terms cancel. $\square$

For the filled tetrahedron Δ^3 , use vertices $(0, 1, 2, 3)$, edges $(01, 02, 03, 12, 13, 23)$, faces $(012, 013, 023, 123)$, and cell 0123. Then

$$(\dim C_0, \dim C_1, \dim C_2, \dim C_3) = (4, 6, 4, 1).$$

The explicit incidence matrices appear in Appendix A. Their ranks are $(3, 3, 1)$. The homology dimensions are consequently $(1, 0, 0, 0)$. This is the filled tetrahedron, not its boundary sphere and not just its edge graph; confusing those carriers changes homology and the operator.

4.2 Closed linear flow

Definition D4.2. On $\mathcal{C}(K)$, define

$$\mathcal{A}_K = \begin{pmatrix} 0 & -B_1 & 0 & 0 \\ B_1^T & 0 & -B_2 & 0 \\ 0 & B_2^T & 0 & -B_3 \\ 0 & 0 & B_3^T & 0 \end{pmatrix}, \quad S_\mu(c) = c^T c.$$

The closed flow is $\dot{c} = \mathcal{A}_K c$, with dimensionless real parameter s .

Lemma T4.3 (standard skew-flow conservation). The generator is skew, and every solution satisfies $dS_\mu/ds = 0$.

Proof. Transposing the block matrix gives $\mathcal{A}_K^T = -\mathcal{A}_K$. Therefore $d(c^T c)/ds = c^T (\mathcal{A}_K^T + \mathcal{A}_K) c = 0$. Equivalently, $e^{s\mathcal{A}_K}$ is orthogonal. $\square$

Let d denote the block map with B_k^T in the degree-raising positions. Then $\mathcal{A} = d - d^T$, while $D = d + d^T$ is the usual symmetric Hodge-Dirac operator. Since $d^2 = 0$,

$$-\mathcal{A}^2 = D^2 = L,$$

where L is block diagonal with

$$L_k = B_k^T B_k + B_{k+1} B_{k+1}^T,$$

using zero terms at the ends. These identities are valid for the declared Euclidean metric. Changing weights requires changing the adjoints consistently.

4.3 The single tetrahedron

Proposition T4.4 (single-cell identity). Let P_0 project onto the uniform vertex vector and vanish on all higher degrees. For the filled tetrahedron,

$$\mathcal{A}^2 = -4(I - P_0), \quad \mathcal{A}^3 + 4\mathcal{A} = 0,$$

and

$$\det(\lambda I - \mathcal{A}) = \lambda(\lambda^2 + 4)^7.$$

Proof. Direct multiplication of Appendix A gives

$$L_0 = 4I_4 - \mathbf{1}\mathbf{1}^T, \quad L_1 = 4I_6, \quad L_2 = 4I_4, \quad L_3 = 4I_1.$$

Thus $L = 4(I - P_0)$ and $\mathcal{A}P_0 = 0$. On the fourteen-dimensional orthogonal complement, $\mathcal{A}^2 = -4I$. A real skew matrix has conjugate eigenvalues in pairs, giving seven copies of each of $2i$ and $-2i$, and one zero eigenvalue. $\square$

The flow can be written without numerical diagonalization:

$$e^{s\mathcal{A}} = P_0 + \cos(2s)(I - P_0) + \frac{\sin(2s)}{2}\mathcal{A}.$$

It has full-operator period π . In clock notation, $s = \pi$ gives phase $2s = 1(2\pi) + 0$, equivalently $2s \equiv 0 \pmod{2\pi}$. This is a periodic parameterization of a matrix flow, not a physical clock.

Proposition P4.5 (orthogonal edge decomposition). On $C_1(\Delta^3)$,

$$P_{\text{cut}} = \frac{1}{4}B_1^T B_1, \quad P_{\text{cycle}} = \frac{1}{4}B_2 B_2^T$$

are orthogonal rank-three projectors, with sum I_6 . In particular, $\ker B_1 = \text{im } B_2$.

Proof. The degree-one block of L is $4I_6$. The two summands have zero product by $B_1 B_2 = 0$; their sum is $4I_6$. Squaring the sum relation after multiplying by either summand shows each squared summand is four times itself. Their ranks are the ranks of B_1 and B_2 , both three. The chain identity gives inclusion of the image in the kernel, and dimensions give equality. $\square$

For a unit vector $z \in \ker B_1$ initially placed in degree one, the solution has

$$c_1(s) = \cos(2s)z, \quad c_2(s) = \frac{1}{2} \sin(2s)B_2^T z, \quad c_0(s) = c_3(s) = 0.$$

This example describes exchange between two chain degrees. It is not transport of a material substance.

5. Incidence balance and unique-simplex gluing

5.1 A local quadratic balance

For incident simplices $\sigma \subset \eta$ of adjacent degree, define the bilinear incidence current

$$J_{\sigma\eta} = 2B_{\sigma\eta}c_\sigma c_\eta.$$

A positive value is directed from the lower degree σ to the upper degree η . This sign convention follows the chosen $\mathcal{A}$; it is not an extra direction assigned to the geometric edge.

For any set R of simplex coordinates, write $S_R = \sum_{\sigma \in R} c_\sigma^2$. The set need not be a subcomplex.

Proposition T5.1 (exact cut balance). Along the closed flow,

$$\frac{dS_R}{ds} = J_{\text{in}}(R) - J_{\text{out}}(R),$$

where only adjacent-degree incidences crossing R and its complement contribute.

Proof. The pair (σ, η) contributes $-2B_{\sigma\eta}c_\sigma c_\eta$ to the derivative of c_σ^2 and the opposite amount to that of c_η^2 . If both coordinates belong to R , these cancel; if neither belongs to R , they do not enter. A crossing incidence contributes with the stated incoming or outgoing sign. Summing over pairs proves the identity. $\square$

The global conservation lemma is the special case containing every coordinate. The local statement additionally identifies which bilinear terms account for a change in a selected sum. It makes no assertion about a physical current observable.

5.2 Shared faces are not duplicated degrees of freedom

For a pure tetrahedral complex, let $n_3(\sigma)$ be the number of top-dimensional tetrahedra containing simplex σ . It is positive for every simplex in this setting. Define an equal ownership weight

$$w_{\sigma\tau} = \begin{cases} 1/n_3(\sigma), & \sigma \subseteq \tau, \\ 0, & \text{otherwise,} \end{cases}$$

and a cell-attributed norm

$$S_\tau^{\text{eq}} = \sum_{\sigma} w_{\sigma\tau} c_\sigma^2.$$

Proposition T5.2 (partition of global support). One has

$$\sum_{\tau} S_\tau^{\text{eq}} = S_\mu(c).$$

Proof. For each unique simplex, the weights of its containing tetrahedra sum to $n_3(\sigma)/n_3(\sigma) = 1$. Interchange the two finite sums. $\square$

For two tetrahedra sharing a face, the three vertices, three edges and that face are common coordinates. They are not copied into independent local chain spaces when forming the global complex. Equal ownership distributes their contribution for accounting, but does not alter the global incidence matrices.

Other nonnegative ownership weights summing to one also give a partition. Equal ownership is a convention, not a derived constitutive law. A naive sum of unweighted cell norms double-counts shared coordinates and does not equal the unique-simplex norm. The theorem requires purity only so the displayed denominator is defined; lower-dimensional components would need a separate attribution rule.

5.3 Gluing, frontier and reduction contracts

The ownership identity concerns a unique-simplex carrier. In a direct sum of independent tetrahedral copies, a shared geometric face instead has separate coordinates in each copy. Passing between these descriptions requires an explicit identification or observation map; the identity in T5.2 is not a dynamical gluing prescription. Unequal prepared stocks do not become equal merely because cells meet along a face.

The archive's frontier and facet reductions are conditional extensions of this distinction. For a regular five-vertex simplex frame, the scale $2/\sqrt{5}$ in a normalized tetrahedral facet reduction follows from the declared frame metric. Choosing a facet, a residual channel and a transfer law does not follow from that scalar. The E2 and facet-residual proposals remain candidates until their selector, metric, boundary and nonlinear compatibility contracts are supplied. Paper III treats shape and growth on prepared carriers; it does not close this selection problem.

6. A finite retriangulation family

6.1 The oriented 2-to-3 identity

On five ordered vertices, use formal oriented 3-chains

$$K_- = [0123] - [0124], \quad K_+ = -[1234] + [0234] - [0134].$$

The first consists of two tetrahedra sharing face 012; the second consists of three sharing edge 34.

Proposition T6.1 (equal exterior boundary).

$$K_- - K_+ = \partial[01234], \quad \partial K_- = \partial K_+.$$

Proof. Expanding the alternating boundary of $[01234]$ gives

$$[1234] - [0234] + [0134] - [0124] + [0123],$$

which is the difference above. Apply $\partial^2 = 0$. $\square$

This is a chain identity. For a legal geometric Pachner move, the link and missing-simplex conditions of the bistellar definition must also hold. The equality does not authorize arbitrary changes in any ambient complex. In particular, it does not identify the two interior operators or give a canonical dynamical interpolation between them.

6.2 The three specified complexes

Let each list below generate its downward-closed simplicial complex:

$$A = \langle 0123, 0124 \rangle, \quad B = \langle 0123, 0124, 0134 \rangle, \quad C = \langle 0134, 0234, 1234 \rangle.$$

Their numbers of unique simplices are

$$f(A) = (5, 9, 7, 2), \quad f(B) = f(C) = (5, 10, 9, 3).$$

Thus $\dim \mathcal{C}(A) = 23$ and $\dim \mathcal{C}(B) = \dim \mathcal{C}(C) = 27$. The intermediate complex B is a declared comparison object, not a claim that a Pachner event physically traverses that state.

Proposition T6.2 (signed B/C isomorphism). The vertex permutation $p = (0\ 3)(1\ 4)$, with vertex 2 fixed, maps B to C . Its degree-wise signed simplex permutation P satisfies

$$P^T P = I, \quad \mathcal{A}_C P = P \mathcal{A}_B.$$

Proof. Apply p to the three maximal simplices and sort each image; the resulting sets are the three maximal simplices of C . The sign of each sorting permutation supplies the oriented chain map. Boundaries commute with this map. Since it is a signed permutation, its inverse is its transpose; the adjoint boundary maps commute as well. Assemble the blocks. $\square$

Proposition P6.3 (inclusion is not flow equivalence). The coordinate inclusion $T : \mathcal{C}(A) \rightarrow \mathcal{C}(B)$, zero on the new coordinates, satisfies $T^T T = I$, but

$$\mathcal{A}_B T - T \mathcal{A}_A \neq 0.$$

Proof. Distinct old simplex basis vectors remain distinct orthonormal vectors, proving isometry. For a concrete residual, vertex 3 is in A , but edge 34 is new in B . The degree-raising block sends the vertex-3 coefficient into the new edge with coefficient -1 , while the included A-flow has no such edge coordinate. Therefore the residual is nonzero. $\square$

This example prevents an event-level norm identity from being mistaken for a continuous-time intertwining theorem. The B/C relabeling has the stronger property; the A/B inclusion does not. Both B and C have complete edge graph on five vertices, but neither is the full clique complex of that graph. Their listed maximal tetrahedra determine which faces and higher simplices exist. Completing every clique would create a different carrier and invalidate the displayed dimensions and spectra. The input is the simplicial complex, not only its one-skeleton.

6.3 Exact spectra

Theorem T6.4 (declared A/B/C spectra). With unique-simplex bases and the unit inner products of Section 4,

$$\begin{aligned} \chi_A(\lambda) &= \lambda(\lambda^2 + 3)^4(\lambda^2 + 5)^7, \\ \chi_B(\lambda) &= \chi_C(\lambda) = \lambda(\lambda^2 + 2)^2(\lambda^2 + 5)^{11}. \end{aligned}$$

Proof by exact matrix certificate. Construct the three incidence systems using the deletion rule in Section 4.1 and Appendix B. Set $L_K = -\mathcal{A}_K^2$. Integer matrices and rational row reduction give the following nullities:

Complex	$\dim \ker L$	$\dim \ker(L - 2I)$	$\dim \ker(L - 3I)$	$\dim \ker(L - 5I)$
A	1	0	8	14
B	1	4	0	22
C	1	4	0	22

In each case the listed nonzero eigenspaces and kernel exhaust the chain dimension. The symmetric matrix L_K has no further eigenvalues. Since $\mathcal{A}_K$ is real skew, a positive eigenvalue ν of L_K splits equally into $\pm i\sqrt{\nu}$ for $\mathcal{A}_K$. This yields the stated characteristic polynomials. The archived matrices and exact row-reduction implementation reproduce every nullity. $\square$

This is a finite certificate rather than a numerical eigenvalue fit. In particular, no floating-point tolerance is used to distinguish zero, two, three or five. The proof depends on the actual complex, not merely on its numbers of simplices. An isomorphic orientation convention gives conjugate matrices and the same result.

No-go NG6.5 (fixed binary 2:2 labels). No fixed binary labeling of the five vertices gives exactly two ones in every four-vertex tetrahedron on this set.

Proof. Write the total number of ones as N , with label b_i at vertex i . The five conditions are $N - b_i = 2$. Thus all b_i are equal. If all are zero, each four-set has zero ones; if all are one, it has four. Both contradict two. $\square$

This excludes one particular auxiliary labeling condition. It does not obstruct the Pachner move itself or rule out another state representation.

6.4 Exact recurrence and its limits

No-go NG6.6 (no positive full-flow period). For each $K \in \{A, B, C\}$, there is no $T > 0$ for which $e^{T\mathcal{A}_K} = I$.

Proof. For A , a full period requires integers $n, m > 0$ with

$$T\sqrt{3} = 2\pi n, \quad T\sqrt{5} = 2\pi m.$$

Thus $\sqrt{5/3} = m/n$. If this were rational in lowest terms, $3m^2 = 5n^2$, contradicting the even valuations of squares at primes 3 and 5. The same argument using primes 2 and 5 rules out $\sqrt{5/2}$ for B and C. $\square$

The clock form of the required equalities is $T\sqrt{3} \equiv T\sqrt{5} \equiv 0 \pmod{2\pi}$ for A, and analogously for B/C: both spectral pointers would have to return to zero after integer full rotations. The irrational ratio forbids a common nonzero exact return.

This statement is about the identity of the entire evolution operator. A harmonic state is fixed. A state supported in a single frequency subspace is periodic. Approximate returns are not excluded; simultaneous rational approximation of phases can bring a bounded finite-dimensional orthogonal flow close to its initial operator. We make no claim about a minimum approximate-return time. The single-cell period π therefore cannot be assigned to all glued complexes, but recurrence of selected states remains possible.

6.5 Ambient comparison and incompatible linear jump contracts

Proposition I.J1 (typed jump obstruction). The two-cell and three-cell sides of the oriented 2-to-3 move have native chain dimensions 23 and 27. Their union has 30 nonempty-simplex coordinates. On that common ambient space distinguish grade-preserving chain compatibility T ,

Euclidean isometry I , pointwise boundary fixation B , all-incidence sign preservation S , and vertex-edge sign preservation S_{01} . Neither $T + I + B$ nor $T + I + S_{01}$ is feasible under the source audit's linear contracts.

Proof of the first obstruction. Boundary fixation and chain compatibility require the source top chain

$$t_- = e_{0123} - e_{0124}$$

to map to the target filling

$$t_+ = -e_{0134} + e_{0234} - e_{1234}.$$

Their squared Euclidean norms are respectively 2 and 3. An isometry cannot implement that forced image. For the second obstruction, the source's constrained face-filling minimization has minimum squared norm $3/2$ for a unit-norm source; hence it also contradicts isometry. This second minimum is an exact finite audit result, with its constraint matrices retained in the source snapshot. The first norm witness is a self-contained proof. $\square$

The full linear chain-map space has affine dimension 96; adding boundary fixation leaves affine dimension 3 in the declared parameterization. These are source-audited solution-space dimensions, not the dimensions of the carriers. Chain compatibility together with all-incidence sign preservation is also infeasible: a deleted face cannot feed the required edge incidences through new target faces. By contrast, some algebraic chain isometries exist when boundary fixation is dropped, and some boundary-fixing coordinate injections exist when chain compatibility is dropped. None selects the native jump. Nonlinear or state-restricted alternatives remain separate questions. The common ambient operator update has rank 18 in the supplied matrices; this rank is not an event frequency or a physical degree count.

7. Recorded events determine affine binary maps

7.1 What data are supplied

Fix pre- and post-event labeled states (g^-, θ^-) and (g^+, θ^+) . Define the relative vertex permutation and binary offset

$$r = g^+(g^-)^{-1}, \quad a = \theta^+ \oplus P_{\pi(r)}\theta^-.$$

The pair (r, a) is the recorded event data. If those endpoints are observed only after the event, the pair is retrospective. To predict an event, a separate pre-event process must supply (r, a) without using the held-out endpoint.

Definition D7.1. For supplied $r \in S_4$, $a \in \Theta$, define the affine map

$$F_{r,a}(\theta) = a \oplus P_{\pi(r)}\theta,$$

and its permutation operator $U_{r,a}e_\theta = e_{F_{r,a}(\theta)}$.

Proposition T7.2 (exact receipt-conditioned map). The map is a bijection, its permutation operator is orthogonal, and endpoint-derived data satisfy $F_{r,a}(\theta^-) = \theta^+$.

Proof. Coordinate permutation and translation in $\mathbb{F}_2^3$ are both bijections. The inverse is $\theta \mapsto P_{\pi(r)}^{-1}(\theta \oplus a)$. A bijection of an orthonormal basis gives an orthogonal matrix. Substituting the endpoint definition of a cancels the repeated binary vector. $\square$

For consecutive supplied events, the maps compose as

$$F_{r_2,a_2} \circ F_{r_1,a_1} = F_{r_2 r_1, a_2 \oplus P_{\pi(r_2)} a_1}.$$

This is an affine-group identity. It does not say which events occur or assign probabilities to them.

7.2 Exact Hamming-sector consequences

Theorem T7.3 (sector transition counts). Let $h = w_H(a)$. For weights w, w' ,

$$\text{rank}(P_{w'} U_{r,a} P_w) = \binom{h}{t} \binom{3-h}{w-t}, \quad t = \frac{w + h - w'}{2},$$

when t is an integer satisfying $0 \leq t \leq h$ and $0 \leq w - t \leq 3 - h$; otherwise the rank is zero. Furthermore,

$$\frac{1}{8} \| [U_{r,a}, W] \|_F^2 = h.$$

Proof. Coordinate permutation preserves weight. Of the h coordinates flipped by the offset, let t initially contain a one. The output weight is $w + h - 2t$. There are $\binom{h}{t} \binom{3-h}{w-t}$ input basis vectors with these properties. Distinct inputs have distinct outputs, so counting them gives the block rank.

For the commutator, its squared norm is the sum of squared weight changes over the eight basis vectors. For a uniformly selected binary vector, each flipped coordinate contributes independently either $+1$ or -1 , equally often. The expected square of their sum is h , since the cross terms cancel. Multiply by eight. $\square$

Thus a map may mix Hamming sectors while preserving the Euclidean norm of every vector on the eight-state space. For example, a one-bit flip has normalized commutator square one but remains a permutation matrix. Hamming-sector preservation for this affine family occurs exactly when $a = 0$.

7.3 Recovering a relative frame when an edge map is supplied

A specified edge map can sometimes determine the vertex permutation in the receipt without reference to the post-event binary endpoint. This remains a conditional extraction problem: the edge map itself must already be available.

Proposition P7.4 (unique relative-frame extraction). Let $M : C_1(\Delta^3) \rightarrow C_1(\Delta^3)$ be a supplied real matrix, and let $J_4 = \mathbf{1}\mathbf{1}^T$. Form

$$\widehat{P} = \frac{1}{4} (B_1 M B_1^T + J_4).$$

If $\widehat{P}$ is a vertex permutation matrix and $B_1 M = \widehat{P} B_1$, there is a unique $r \in S_4$ with $P_0(r) = \widehat{P}$. With the induced signed edge map $P_1(r)$, the residual $R = P_1(r)^{-1} M$ satisfies $B_1 R = B_1$. If M is orthogonal, R fixes the cut space pointwise and preserves the cycle space.

Proof. A permutation matrix uniquely specifies its vertex permutation. Boundary equivariance gives $B_1 P_1(r) = P_0(r) B_1$, so substitution gives $B_1 R = B_1$. Transposing yields $R^T B_1^T = B_1^T$. If R is orthogonal, multiply by R to obtain $R B_1^T = B_1^T$; hence it fixes the cut space. Orthogonality then preserves its orthogonal complement, the cycle space. Conversely, if an edge map satisfies $B_1 M = P_0 B_1$ for a permutation P_0 , multiplication by B_1^T and use of $B_1 B_1^T = 4I - J_4$, $P_0 J_4 = J_4$, recovers the stated formula. $\square$

The two admission checks are essential. The formula alone may yield a matrix that is not a permutation, and a visually nearby permutation is not an exact substitute. Even successful extraction leaves the parity offset a unspecified. This illustrates the distinction between recovering a frame from sufficient recorded data and deriving an event source.

7.4 The unresolved source question

The logical conclusion is deliberately conditional:

$$\text{supplied } (r, a) \implies \text{an exact affine } \Theta\text{-map.}$$

It is not a derivation of (r, a) from the chain state or from a microscopic law. Extracting an offset from observed endpoints always fits that pair of endpoints by definition; it cannot count as a successful prediction of them. Many different offset-selection rules are compatible with the same carrier, group action and closed norm conservation.

Later conditional constructions in the research archive introduce finite update interfaces, source packets, recovery criteria and event-admission rules. They specify what a source would have to provide. They are not used as premises of the theorems above and are not stacked into the main argument. In particular, an executable rule that accepts source terms as input is not a derivation of those terms. The source of predictive receipts remains outside this paper.

7.5 Normalization, masks and historical constitutive proposals

A supplied two-rate response can be parameterized as $k_+ = \gamma\kappa$, $k_- = \gamma/\kappa$, with $\gamma = \sqrt{k_+k_-}$ and $\kappa = \sqrt{k_+/k_-}$. This is a change of variables for positive rates. It does not determine either rate from incidence. If a coordinate transformation is N , its induced metric is $N^T N$; using the raw Euclidean norm in both coordinate systems without that transformation changes the assertion. The parity, cotangent and response-Hessian addenda are consolidated under these declared-map and declared-metric conditions.

An idle mask and an intact null event can produce the same visible endpoint data. Receipt extraction after an event therefore does not supply a predictive pre-event source. A receipt law must state what fixes the frame permutation, binary offset, mask, admission and timing before execution. The archived winding, holonomy, ring and chirality developments likewise retain their linear or quadratic source forks and supplied source amplitudes. The historical seed value 101/6 is not a final particle burden. Rank-two frame holonomy and the recorded $n = 1, 2$ closures are conditional algebra on their specified carriers, with no inferred material identity or physical burden unit.

8. Signed readiness and supplied contact dynamics

8.1 Fixed coupling and an empty prepared recipient

Let $x, y \in \mathbb{R}^{15}$ be independent filled-tetrahedron copies, with

$$x' = \mathcal{A}x + By, \quad y' = \mathcal{A}y + Bx, \quad B = aB_{01} + bB_{12}, \quad B^T = -B.$$

The face-supported skew channels B_{01}, B_{12} are those in the readiness source package, distinct from the simplicial boundary matrices. This notation must not identify B_{01} with B_1 . Their support excludes the harmonic vertex mode, so $B\mathcal{A}^2 = -4B$.

Theorem I.R1 (fixed-coupling readiness). For $x(0) = v, y(0) = 0$: if $Bv \neq 0$, recipient stock starts as $\|Bv\|^2 s^2 + O(s^3)$; if $Bv = 0$ but $B\mathcal{A}v \neq 0$, it starts as $\|B\mathcal{A}v\|^2 s^4/4 + O(s^5)$. If both vectors vanish, the recipient remains zero for all time.

Proof. Differentiate the second equation at zero: $y'(0) = Bv$. When this vanishes, $y''(0) = B\mathcal{A}v$. Taylor expansion gives the two leading squared norms. If both vanish, $B\mathcal{A}^2 = -4B$ makes

$\ker B \cap \ker BA$ invariant under $\mathcal{A}$. Thus $(e^{s\mathcal{A}}v, 0)$ solves the coupled equations, and uniqueness proves the last assertion. $\square$

Equal stocks do not specify readiness. The source's three equal-stock positive spokes are silent, whereas changing one amplitude sign gives leading recipient stock $2a^2s^4$. The recipient already exists in the direct-sum carrier: its recruitment is population of existing coordinates, not creation of a tetrahedron. A nonlinear overlap rule whose two coefficients vanish whenever the recipient is empty leaves that empty-copy set invariant. It cannot be used as evidence of spontaneous recruitment.

8.2 A seed-capable candidate and its supplied records

For the seven-coordinate shared-face projector F , the audited channels satisfy $B_j^T B_j = 3P_j$, with orthogonal projectors P_{01}, P_{12} and $P_{01} + P_{12} \preceq F$. Define

$$T_F = \|Fx\|^2 + \|Fy\|^2, \quad R_j = (\|B_j x\|^2 + \|B_j y\|^2)/3, \quad g_j = R_j/T_F$$

when $T_F > 0$, and set $g_j = 0$ otherwise. Then $g_j \geq 0$ and $g_{01} + g_{12} \leq 1$. The candidate coupling is $\chi(g_{01}B_{01} + g_{12}B_{12})$, with a supplied polarity $\chi \in \{\pm 1\}$. This quotient and the identity response are design choices. The source proves global continuation and norm conservation for this candidate; they do not select it as native.

The polarity is especially consequential at an empty recipient. Whole-copy amplitude reversal fixes an empty input yet reverses the required polarity transformation. An amplitude-only covariant polarity selector would therefore require $\chi = -\chi$, impossible for $\chi = \pm 1$. A reference, history or prepared record must be supplied. State-dependent skewness conserves the single solution's total stock; it does not in general preserve distances between different solutions.

8.3 Multi-copy charts, currents and order

For a supplied graph of independent copies, let $C_{i,e}$ be the orthogonal endpoint chart at contact $e = (i, k)$, commuting with the local $\mathcal{A}$. In these charts set

$$K_{ik} = C_{i,e}(g_{01}B_{01} + g_{12}B_{12})C_{k,e}^T, \quad K_{ki} = -K_{ik}^T.$$

The reverse block must use this transpose rule; different endpoint charts do not permit copying the same numerical block into both positions. Pairwise cancellation then gives total-stock conservation and the reciprocal contact current

$$J_{i \leftarrow k} = 2\chi_e x_i^T K_{ik} x_k = -J_{k \leftarrow i}.$$

For bounded degree Δ , the audited contact Lipschitz bound $4\sqrt{3}$ gives global bound $2 + 4\sqrt{3}\Delta$, and hence unique global continuation. A polarity graph can be transformed to an all-positive convention exactly when every cycle product is positive: propagate vertex signs along a tree, and the remaining edges test precisely those products. Negative-cycle record states are allowed by the candidate; balance is an additional condition.

Disjoint contacts commute. Shared-copy contacts need not. On the legal tetrahedra 0123, 0124, 0135, the central contacts use distinct faces 012, 013. With the audited chart $(0, 1, 3, 2)$ and a prepared face vector, the contact bracket is $(0, 2v, -2v)$, squared norm 16. Thus geometric legality does not make sequential updates confluent. The abstract common-face example has bracket $(3v, 0, -3v)$, squared norm 36; it should not replace the legal distinct-face witness. Prepared three-copy recruitment gives first-recipient stock $6s^2$ and terminal stock $9s^4/2$: higher-order onset is not a finite waiting interval.

8.4 Admission and collective history remain source contracts

No-go I.R2 (symmetric exclusive progress). If a prepared state and geometry are invariant under an automorphism exchanging two eligible contacts, no deterministic covariant selector can choose exactly one of those contacts while always making progress.

Proof. Covariance requires the selected subset to be invariant under the exchange. Its only invariant subsets are neither contact or both. The former violates progress and the latter exclusivity. $\square$

Potential incidence is therefore distinct from the active contact graph. In the source witness, enabling both contacts or neither gives a vector-field difference of squared norm 32 at the same visible geometry and readiness. Prepared opposite polarities at an empty endpoint give derivative difference of squared norm 24. Batch authorization, hold, priority, history or stochastic records are distinct possible extensions. The existing binary-state hold/tie rule cannot automatically be transferred to this amplitude graph.

The four-face collective history proposal supplies local rational/Cayley rotations, complete shared-face observations, amplitude, schedule, launch, step, clock and embedding. Its same-history lift is conditional on those observations. The 53 exact checks, 730 numerical checks and 720 schedule trials are bounded implementation evidence, not a universal propagation or isotropy theorem. The 519 multi-copy checks and 551 arbitration checks with 128 structured fixtures are constructor diagnostics. Independent human review is pending.

9. Limits, physical dictionary and operational checks

The finite carrier, its skew flow, a jump map, readiness coefficients and an admission law remain separately typed objects. This edition integrates progress in each without identifying their source choices. Appendix C retains the earlier distinctions among Gram defect, a second quadratic form, Hamming-sector mixing and complementary effects. A fixed skew flow preserves distances and cannot asymptotically attract an open set to a proper lower-dimensional pattern. The audited nonlinear readiness flow is volume preserving under its stated formula; norm preservation alone would not prove that stronger property.

Action rescaling, a canonical units dictionary and a stipulated clock comparison do not predict $\hbar$, a gravitational coupling or seconds per unit s . Fixed-chain dimensional obstructions and physical-to-Hodge projectability counterexamples remain relevant even when a modal frequency can be rescaled to match. Event time in the journal core v0.3.2 is a counter convention with a SER invariant; it is not an acquired causal metric or LC period. Winding and formation require supplied source amplitudes, material identification and predictive units. The engineering addenda introduce reciprocal energy, activation, damping, sensing, latency and finite control windows as explicit mechanism contracts.

The empirical and collaboration records are preserved in the integration register. Optical/QuiX, LC, calorimetry and external corpus analyses have separate acquisition assumptions. LC Stage 0 remains UNSCORED at the stated threshold 3/200. An external-domain data set or synthetic fixture cannot establish the same-carrier physical bridge. The attached coefficient ledger v0.39 is a dated baseline; later source-authoritative amendments govern the present claim ceiling.

Operational checks. A nonzero boundary square refutes the carrier construction. Unequal declared fibers refute P3.3. A failure of ownership weights to sum to one refutes T5.2. A positive full-operator period would refute NG6.6; a single-state period would not. Failure of endpoint reciprocity refutes the supplied multi-copy conservation argument. Reconstruction after an event

cannot be scored as successful prediction before it. Physical promotion remains zero. # 10. Claim table and reproduction

The table is an index of proof obligations, not a count of scientific merit. “Standard lemma” identifies conventional setup. “Finite calculation” identifies a reproducible specialization. Artifact references point to the supplement’s `data/exact_artifacts.json` and `verify.py`.

ID	Statement	Status	Proof / artifact	Depends on
P3.1	Matching action has kernel V_4	Proposition	Group argument; exhaustive action	Three matchings
P3.2	Matching stabilizer differs from Θ	Proposition	Noncommuting elements	P3.1
P3.3	Six edge fibers each have 32 states	Proposition	Four permutations per edge	Declared p_E
NG3.4	Fiber size does not imply depth	No-go	Three maps on the same set	P3.3
P3.5	Hamming ranks are 1,3,3,1	Proposition	Binary enumeration	W
L4.1	Boundary squared vanishes	Standard lemma	Sign cancellation; matrices	Orientation rule
T4.3	Closed flow conserves S_μ	Standard lemma	Skew identity	D4.2
T4.4	Single-cell squared generator	Finite calculation	Exact block multiplication	Appendix A
P4.5	Edge cut/cycle projectors	Proposition	Projector identities	L4.1, T4.4
T5.1	Cut norm balance	Proposition	Pairwise cancellation	D4.2
T5.2	Ownership sums to global norm	Proposition	Partition of unity	Unique simplices
T6.1	Pachner exterior boundaries agree	Standard identity	Alternating boundary	L4.1
T6.2	Signed B/C intertwiner	Finite calculation	Signed permutation matrix	Specified B/C
P6.3	A/B inclusion is not an intertwiner	Counterexample	New edge 34 residual	Specified A/B
T6.4	Exact A/B/C spectra	Finite theorem	Rational nullity certificates	D4.2
NG6.5	No common binary 2:2 labeling	No-go	Five sum equations	Five vertices
NG6.6	No positive full-flow period	No-go	Irrational frequency ratios	T6.4
T7.2	Supplied receipts give affine bijection	Proposition	Inverse map	D7.1
T7.3	Transition ranks and commutator	Finite theorem	Counting; exhaustive replay	P3.5, D7.1
P7.4	Admitted edge map fixes relative frame	Proposition	Boundary identity; permutations	Supplied edge map
P8.1	Norm and second-form preservation differ	Counterexample	Two-dimensional matrices	Declared Q

Run `python baseline/verify.py` from the extracted reproduction package. It regenerates the ordered simplex bases, incidence matrices, generators, A/B/C maps, nullity certificates, matching actions, affine transition counts and the companion PSR fixtures. Only the Python standard

library is required. The README records the measured execution time; the exact algebra uses integers and rational numbers rather than Monte Carlo or fitted tolerances.

The replay verifies finite algebraic instances and supplied artifacts. The general cancellation, counting and reconstruction arguments are the written proofs. Agreement of software output is not independent mathematical peer review. The package also records file hashes so a reviewer can identify the exact files tested; those hashes are not premises of any theorem.

Additional integrated statements and their review surface:

ID	Statement	Evidence / premise	Location
I.J1	Linear jump contract obstructions	Exact norm witness; source finite audit	6.5
I.R1	Fixed-coupling empty-recipient criterion	Written Taylor/invariance proof	8.1
I.R2	Symmetric exclusive progress impossible	Covariance proof	8.4
I.CAND	Seed-capable reciprocal contact continuation	Supplied coefficients, charts and polarity	8.2-8.3
I.ORDER	Legal distinct-face contact order witness	Source finite audit, squared bracket 16	8.3

Appendix A. Single-cell matrices

With the bases in Section 4,

$$B_1 = \begin{pmatrix} -1 & -1 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix},$$

$$B_2 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad B_3 = \begin{pmatrix} -1 \\ 1 \\ -1 \\ 1 \end{pmatrix}.$$

The vertex block product is

$$B_1 B_1^T = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix}.$$

The uniform vertex projector is $\mathbf{1}\mathbf{1}^T/4$, extended by zero to the other eleven coordinates. Multiplication gives $B_1 B_2 = 0$, $B_2 B_3 = 0$, $B_1^T B_1 + B_2 B_2^T = 4I_6$, $B_2^T B_2 + B_3 B_3^T = 4I_4$, and $B_3^T B_3 = 4$.

An orientation reversal multiplies a simplex basis vector by minus one. Reversing several orientations gives diagonal sign matrices R_k , changes the boundary to $R_{k-1} B_k R_k$, and conjugates the full generator by $\text{diag}(R_0, R_1, R_2, R_3)$. Hence the spectrum and global norm do not depend on this basis choice.

Appendix B. Reconstructing the finite certificates

A reviewer can construct every matrix without any private source or conversation. For each listed maximal tetrahedron, enumerate every nonempty subset. Deduplicate equal vertex sets, sort by dimension and then lexicographically, and orient every simplex by increasing vertex order. Insert the alternating deletion signs into the boundary matrices. This determines the generator without a free matrix entry.

For A, the edge list is

$$01, 02, 03, 04, 12, 13, 14, 23, 24,$$

and the face list is

$$012, 013, 014, 023, 024, 123, 124.$$

For B the extra edge is 34, the extra faces are 034, 134, and the extra tetrahedron is 0134. C follows by the vertex permutation in T6.2; its independent lexicographic construction is also supplied. The exact boundary ranks are $(4, 5, 2)$ for A and $(4, 6, 3)$ for B/C, giving one degree-zero homology class and no higher homology in each case.

The squared-frequency certificate can be checked by Gaussian elimination over rational numbers. For A the ranks of L , $L - 3I$, $L - 5I$ are 22, 15, 9; subtracting from 23 gives 1, 8, 14. For B/C the ranks of L , $L - 2I$, $L - 5I$ are 26, 23, 5, giving 1, 4, 22. The dimensions sum to the full space, so the displayed eigenvalues exhaust a symmetric matrix's spectrum.

One need not approximate square roots to verify the absence of a full period. Only the exact squared frequencies and the elementary irrationality arguments in NG6.6 enter that conclusion. Likewise, one need not integrate the differential equation numerically to verify support conservation.

Appendix C. Four preservation statements with different types

Let T be a linear map between Euclidean chain spaces. Its Gram defect is

$$G_T = T^T T - I.$$

It measures a change in squared chain norm: $\|Tc\|^2 - \|c\|^2 = c^T G_T c$. This must not be confused with Hamming-sector mixing on the separate eight-state space.

Suppose a second quadratic form is specified by a symmetric matrix Q . The change of that form is

$$c^T (T^T Q T - Q) c.$$

A project may call such a second functional *burden*, but the word supplies no physical meaning and no relation to S_μ . The matrix Q and its carrier must be stated before the expression is meaningful.

Proposition P8.1 (norm preservation does not imply preservation of another form).

There is an orthogonal T for which $T^T Q T \neq Q$.

Proof. In two dimensions, let T exchange the coordinates and $Q = \text{diag}(1, 2)$. Then $T^T T = I$, but $T^T Q T = \text{diag}(2, 1) \neq Q$. $\square$

Finally, a contraction K , satisfying $K^T K \preceq I$, has the nonnegative complementary effect $E = I - K^T K$. Choosing $L = E^{1/2}$ gives

$$K^T K + L^T L = I.$$

This is the elementary finite operator identity behind a retained/lost split. In a complex quantum-operation setting it is familiar from Kraus representations. Here it is only linear algebra. A permutation mixing Hamming sectors can have zero Gram defect and zero complementary effect; a contraction can have nonzero complementary effect without any binary-state interpretation.

These examples explain why numerical agreement between four diagnostics would not make them one object. Each has a domain, codomain and defining map. Calling all of them “leakage,” “transport” or “burden” would erase premises needed by a referee to check the claim.

Appendix D. Source integration and evidence authority

This edition consolidates the 26 thematic source families from the M1 inventory. The companion `INTEGRATION_DISPOSITIONS.json` and filterable register contain all 984 routed records (955 registered items and 29 additional addendum paths), with original paths, hashes, disposition and this edition's body/digest anchors. The addendum inventory contains 166 paths and 145 distinct hashes; repeated snapshots are cross-references, not additional evidence. Inventory coverage and editorial integration are not independent certification of every underlying proof. Detailed historical apparatus reports remain linked source evidence rather than new physical claims.

Legacy proposition IDs from Papers I and II edition 1.0 are retained even where a section moved. New IDs carry the paper prefix. Paper III uses the consecutive statement numbers 1-16 of its retained core, followed by the explicitly prefixed new statements. Section numbers and claim-table IDs therefore need not coincide.

A CRL label is a dated governance assertion. The supplied CRL2 subdivision proposal distinguishes proposal, preregistration/review, blinded execution and independent replication; it is not evidence that the last stages occurred. Citation depth does not deepen evidence, and a changed assumption demotes dependent conclusions until reviewed. Candidate amendments remain development-lane proposals. The coefficient ledger v0.39 supplied with older packages is an August baseline, not the ceiling for later amendments.

Original unavailable artifacts are named as provenance gaps. Reconstructed proofs and certificates have their own identity and never stand in for recovered historical bytes. Full portal publication and presentation reconciliation remain a subsequent release task.

Archive authority is determined by bytes and embedded edition together: the v2.33-named checkpoint embeds v2.32, and the v2.40-named checkpoint embeds older v2.36 text. Both names are retained as lineage, without silently treating their filenames as current scientific assertions. The journal core v0.3.2 supersedes the earlier v0.3.1 event-time account for the present digest.

F01. Finite carrier, chain algebra and A/B/C baseline

Body location: 4; 6. **Evidence class:** EXACT / NO-GO. **Source records:** 2.

Retain current proofs; make carrier dimensions, orientation, unit adjoint, event inclusion and continuous intertwining explicit. Attach the 1119-assertion reproduction receipt.

Remaining limit: No new fixed-carrier theorem needed; preserve the A-to-B nonintertwining and recurrence obstructions.

- Source record **manuscript**: Paper I - Finite Carrier and Hodge-Dirac Support. Original path and SHA-256 are retained in the companion register.
- Source record **reviewClaims**: Paper I - Claim Table. Original path and SHA-256 are retained in the companion register.

F02. Ambient bistellar comparison and linear jump contracts

Body location: 6.5. **Evidence class:** EXACT / NO-GO / OPEN. **Source records:** 16.

Add native dimensions 23/27 and ambient dimension 30; distinguish overlap, chain compatibility T, isometry I, boundary fixation B, all-incidence signs S and S01. Include rank-18 update, affine dimension classifications and incompatibility witnesses.

Remaining limit: Do not select a new jump law from an existence certificate; preserve nonlinear and selected-state alternatives as open.

- Source record **s29Interface**: Geometric inner-product interface. Original path and SHA-256 are retained in the companion register.
- Source record **s29InterfaceResults**: Geometric inner-product interface - exact results. Original path and SHA-256 are retained in the companion register.
- Source record **s29InterfaceVerifier**: Geometric inner-product interface - verifier. Original path and SHA-256 are retained in the companion register.

F03. Gluing, frontier, facet reduction and E2

Body location: 5.3. **Evidence class:** EXACT / CONDITIONAL / OPEN. **Source records:** 71.

Consolidate unique-simplex accounting, unequal support transfer, Delta4-to-K4 reduction and facet residual candidates. Distinguish the forced scale $2/\sqrt{5}$ from a supplied selector and physical transfer.

Remaining limit: Facet selection, nonlinear compatibility and E2 physical qualification remain separate obligations.

- Source record **frontierTransfer**: Common-Face Transfer and Unequal Channels. Original path and SHA-256 are retained in the companion register.
- Source record **frontierAccessibility**: Coarse Current and Obstruction Observables. Original path and SHA-256 are retained in the companion register.
- Source record **frontierContrast**: Support Contrast under Multiplicative Transfer. Original path and SHA-256 are retained in the companion register.

F04. Parity, response Hessians and normalization

Body location: 7.5. **Evidence class:** EXACT / CONDITIONAL / CANDIDATE. **Source records:** 138.

Collect fixed-event involution, cotangent/action normalization, parity descent and response-Hessian alignment. State each ambient space and response law; retain CRL claims as dated governance assertions.

Remaining limit: Method replay does not certify a new primitive, action unit or independent review.

- Source record **parityExecutionSource1**: Crl2D And Governance Review Status. Original path and SHA-256 are retained in the companion register.
- Source record **parityExecutionSource2**: Crl2D Promotion And Governance Review Status. Original path and SHA-256 are retained in the companion register.
- Source record **parityExecutionSource3**: Event Hodge Parity Exact Checks. Original path and SHA-256 are retained in the companion register.

F05. Constitutive receipts, masks and pre-event source

Body location: 7.4-7.5. **Evidence class:** CONDITIONAL / NO-GO / OPEN. **Source records:** 54.

Keep the affine event map after receipts are supplied; add idle-mask and intact-null ambiguity, source extraction and pre-event nonidentifiability. Separate reconstruction after an event from prediction before it.

Remaining limit: A source-complete predictive receipt and acquisition map are still required.

- Source record AD003: UD JOP C1 HISTORICAL EDGE AMPLITUDE SEED MINING ADDENDUM v0.1. Original path and SHA-256 are retained in the companion register.

F06. Signed seed readiness and local activation

Body location: 8.1-8.2. **Evidence class:** EXACT / CONDITIONAL / CANDIDATE / NO-GO. **Source records:** 6.

Add $B_v = B_{Av} = 0$ criterion, signed equal-stock counterexample, prepared-polarity seed-capable continuation, and failed overlap recruitment. Keep fixed and nonlinear coefficient contracts distinct.

Remaining limit: Existing-copy population is not geometric carrier creation; native polarity, activation and binding remain open.

- Source record AD139: UD MANUSCRIPT ADDENDUM LOCAL ACTIVATION AND SEED v0.1. Original path and SHA-256 are retained in the companion register.
- Source record AD140: SOURCE ADDENDUM CANDIDATE FACE READINESS AND POLARITY v0.1. Original path and SHA-256 are retained in the companion register.
- Source record AD141: UD MANUSCRIPT ADDENDUM SEED READINESS v0.1. Original path and SHA-256 are retained in the companion register.

F07. Multi-copy readiness and contact arbitration

Body location: 8.3-8.4. **Evidence class:** CONDITIONAL / NO-GO / OPEN. **Source records:** 5.

Integrate reciprocal endpoint charts on supplied graphs, global continuation bound, cycle polarity criterion, order brackets and legal distinct-face example. Add admission versus potential incidence distinction and symmetric exclusive-selector no-go.

Remaining limit: 551 exact assertions/128 structured graph fixtures and 519 checks are constructor diagnostics; native admission, collective authorization and independent review remain open.

- Source record AD106: SOURCE ADDENDUM CANDIDATE FACE READINESS AND POLARITY v0.1. Original path and SHA-256 are retained in the companion register.

F20. Event time and finite observation

Body location: 9; II 9. **Evidence class:** EXACT event-count convention / OPEN metric bridge. **Source records:** 15.

Cite journal core v0.3.2; summarize SER counter invariant and finite observation/recorder constraints. Preserve earlier full manuscript as lineage and attribute standard event-count constructions.

Remaining limit: Event counts are not seconds, causal metric duration or a measured LC period; journal handoff is not submission-ready peer review.

- Source record `timeBenchSession`: Event time and the next measured LC period. Original path and SHA-256 are retained in the companion register.
- Source record `eventTimeReference`: Event Time and Finite Observation. Original path and SHA-256 are retained in the companion register.
- Source record `stage0Estimator`: K4 Stage 0 peak-estimator addendum. Original path and SHA-256 are retained in the companion register.

F21. Action, clock, units and physical dictionary

Body location: 9; III 14. **Evidence class:** CONDITIONAL / NO-GO / OPEN. **Source records:** 70.

Consolidate action-scale freedom, fixed-chain dimensional obstruction, canonical units, gravity/clock proposal boundaries and physical-to-Hodge projectability counterexamples.

Remaining limit: An internal normalization, engineered comparator or rescaled modal match is not a prediction of $\hbar$, gravity or a physical time scale.

- Source record **AD046:** UD SOURCE ADDENDUM INTERNAL ENERGY v0.1. Original path and SHA-256 are retained in the companion register.
- Source record **AD047:** UD SOURCE ADDENDUM PARITY PARENT v0.1. Original path and SHA-256 are retained in the companion register.

F23. Winding, formation and material topology

Body location: 7.5; III 14. **Evidence class:** CONDITIONAL / NO-GO / OPEN. **Source records:** 49.

Extract enduring structural controls from v2.1-v2.41: rank-2 holonomy, $n=1/n=2$ closure, winding maps, linear/quadratic source forks, translation audit and retained ring/chirality conditions.

Remaining limit: Source amplitudes, physical burden units, material identities and predictive formation laws remain unresolved; historical seed 101/6 is not a final particle burden.

- Source record **retainedTopologyReview:** Retained Relations and Conditional Loop Structure. Original path and SHA-256 are retained in the companion register.
- Source record **windingPreregPacket:** Winding and Response Pre-Registration Packet v1.0. Original path and SHA-256 are retained in the companion register.
- Source record **windingPacketErrata:** Winding/Response Packet v1.1 Errata. Original path and SHA-256 are retained in the companion register.

F25. Governance, release authority and manuscript lineage

Body location: Appendix D; II Appendix B; III Appendix A. **Evidence class:** HISTORICAL / GOVERNANCE / OPEN review. **Source records:** 101.

Retain immutable versioned artifacts, latest source-authoritative statements, dated CRL assertions and archive succession. Correct filename/frontmatter drift and current versus historical labels.

Remaining limit: Constructor execution, AI review and governance labels are not independent peer review; full portal refresh is M3.

- Source record **fundingReplay:** Funding review replay - exact spectra and continuity. Original path and SHA-256 are retained in the companion register.
- Source record **fundingReplayResults:** Funding review replay - exact result receipt. Original path and SHA-256 are retained in the companion register.
- Source record **fundingReplayCode:** Funding review replay - supplemental verifier. Original path and SHA-256 are retained in the companion register.

F26. Collective four-face history lift

Body location: 8.4; II 8.3; III 12.3. **Evidence class:** CONDITIONAL / CANDIDATE / OPEN. **Source records:** 1.

Add local rational/Cayley rotations and same-history lift under complete shared-face observations; record supplied amplitude, schedule, launch, step, clock and embedding.

Remaining limit: 720 schedule tests support bounded implementation behavior, not a universal propagation law or isotropy theorem.

- Source record `r1003CollectiveHistory`: Candidate collective four-face history lift. Original path and SHA-256 are retained in the companion register.

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